

Characterization of Goalkeeper Actions from Skeletal Tracking Data

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Introduction

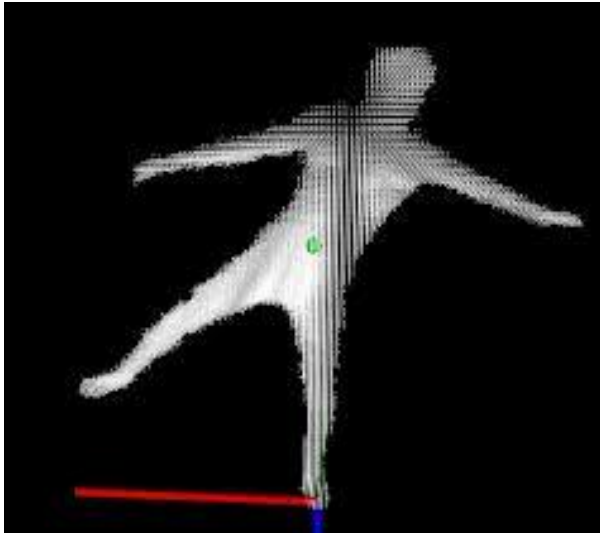


Illustration from T. Kaichi et al., "Estimation of Center of Mass for Sports Scene Using Weighted Visual Hull," 2018 IEEE (CVPRW)

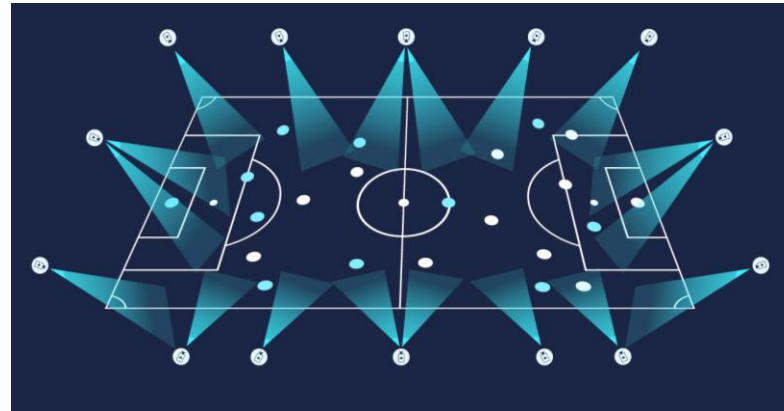


Illustration from Genius Sports

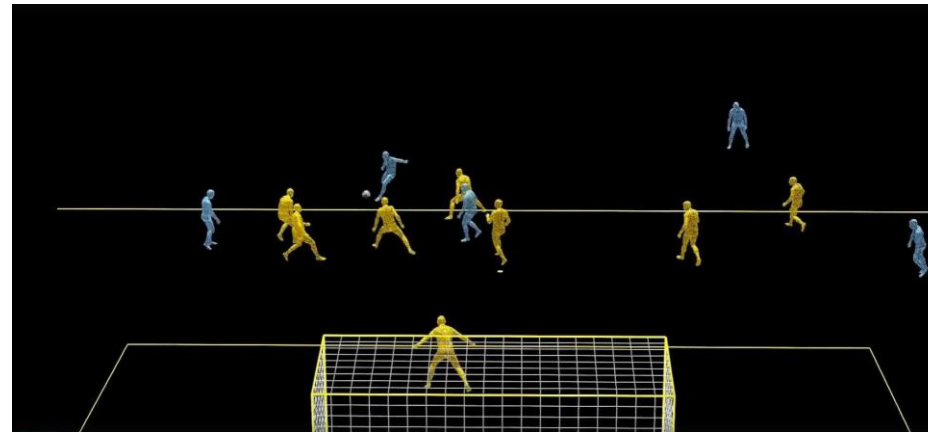


Illustration from Genius Sports

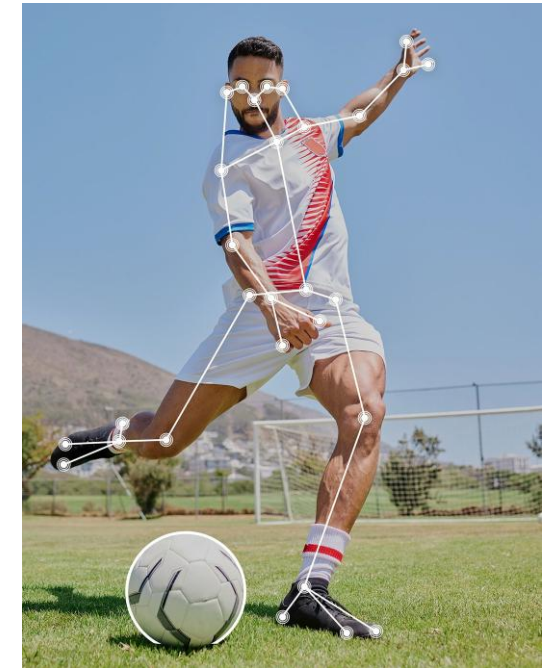
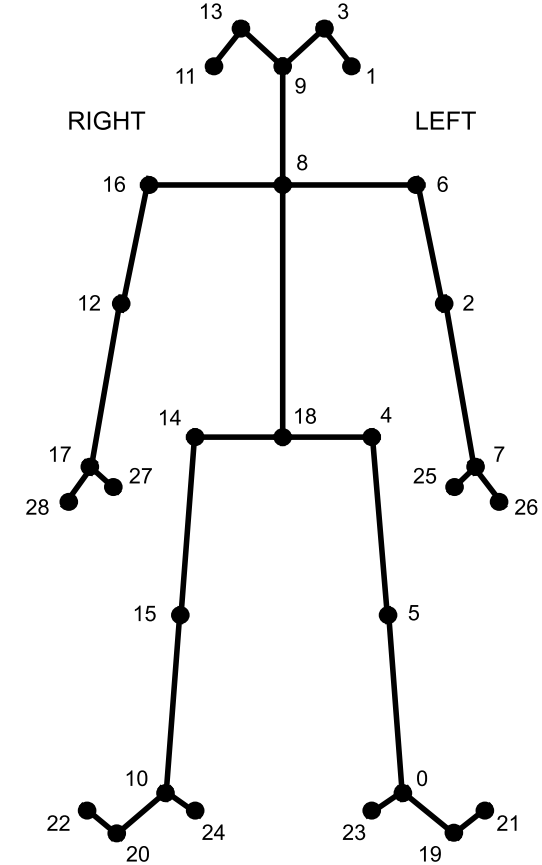


Illustration from Hawk-Eye/Sony

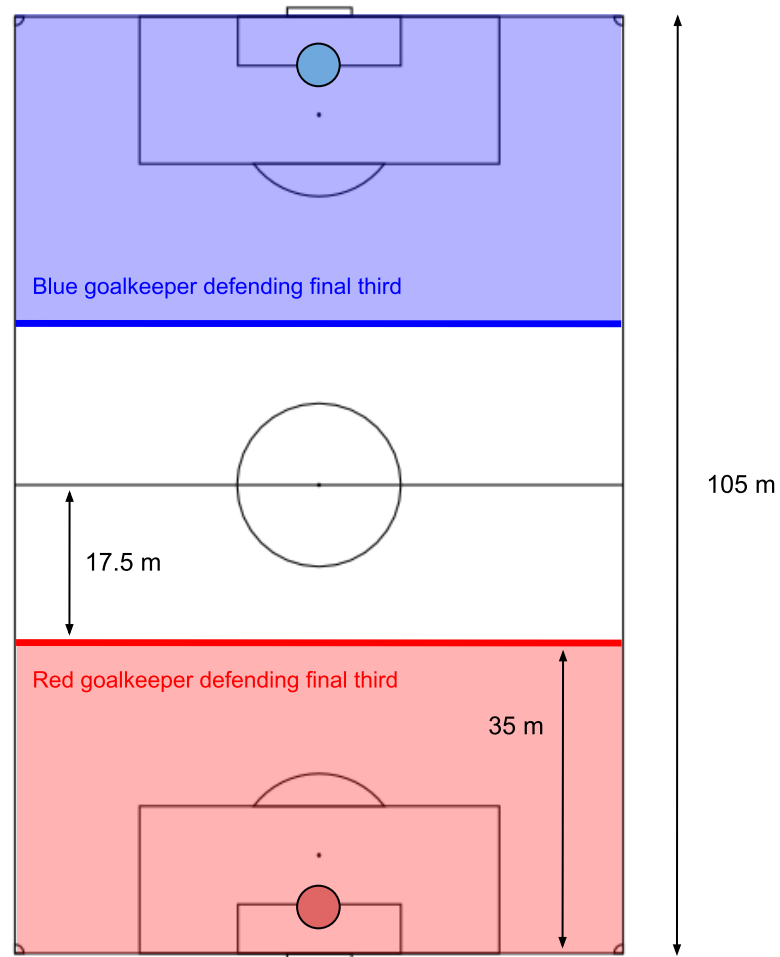
Dataset

- All 51 matches of the UEFA Euro 2024 tournament
- Data contains ball position, player centroids and skeletal data
- Sampled at 25 frames per second
- Skeletal data consists of 3D coordinates (in meters) of all 29 skeletal joints for every player and referee on the pitch
- In total, over 150 GB of raw data (compressed)



0 - Left ankle	8 - Neck	16 - Right shoulder	24 - Right heel
1 - Left ear	9 - Nose	17 - Right wrist	25 - Left thumb
2 - Left elbow	10 - Right ankle	18 - Mid hip	26 - Left pinkie
3 - Left eye	11 - Right ear	19 - Left big-toe	27 - Right thumb
4 - Left hip	12 - Right elbow	20 - Right big-toe	28 - Right pinkie
5 - Left knee	13 - Right eye	21 - Left small-toe	
6 - Left shoulder	14 - Right hip	22 - Right small-toe	
7 - Left wrist	15 - Right knee	23 - Left heel	

Data Preprocessing — Data filtering

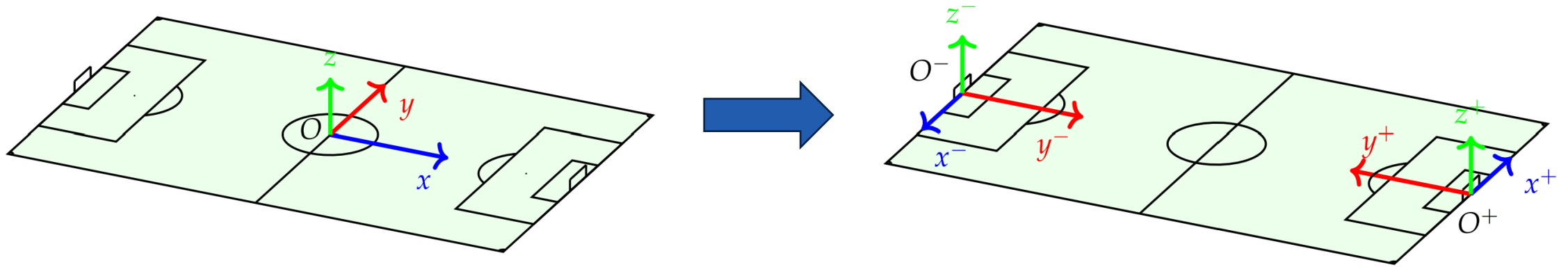


Only keep frames in which:

- Player has role “Goalkeeper”, and
- ball is within the goalkeeper’s final defending third

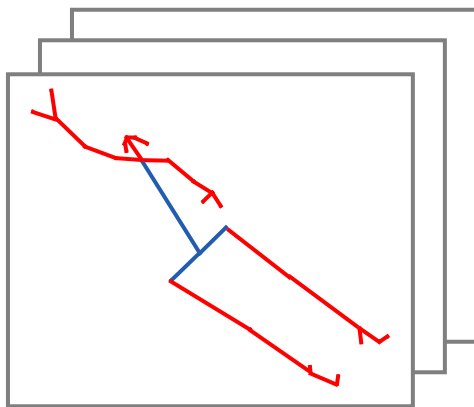
Then, aggregate frames into contiguous sequences per goalkeeper

Data Preprocessing — Goal centering and Body resizing



Pitch-centric coordinates (x, y, z) $\rightarrow$ **Goal-centered coordinates (x^-, y^-, z^-) and (x^+, y^+, z^+)**

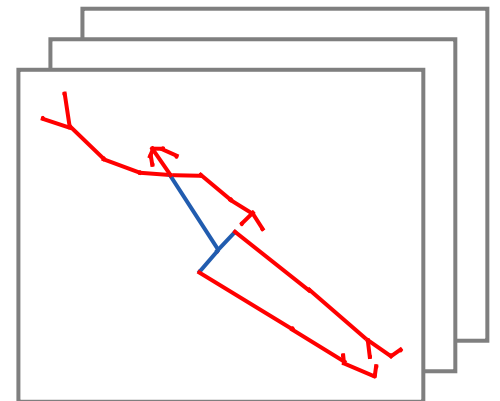
Body-size normalization:



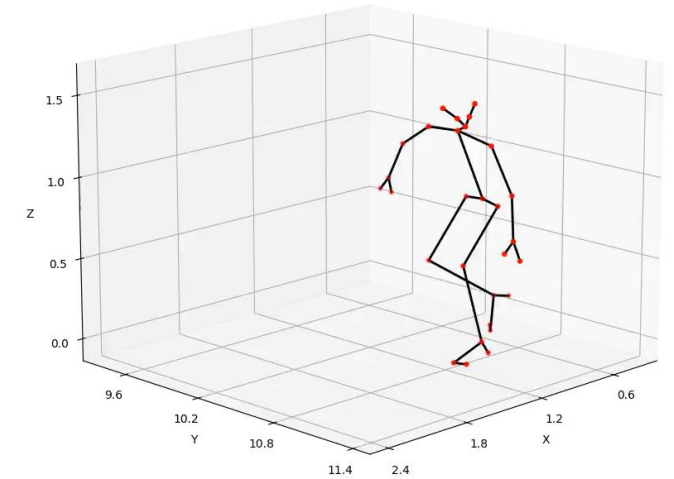
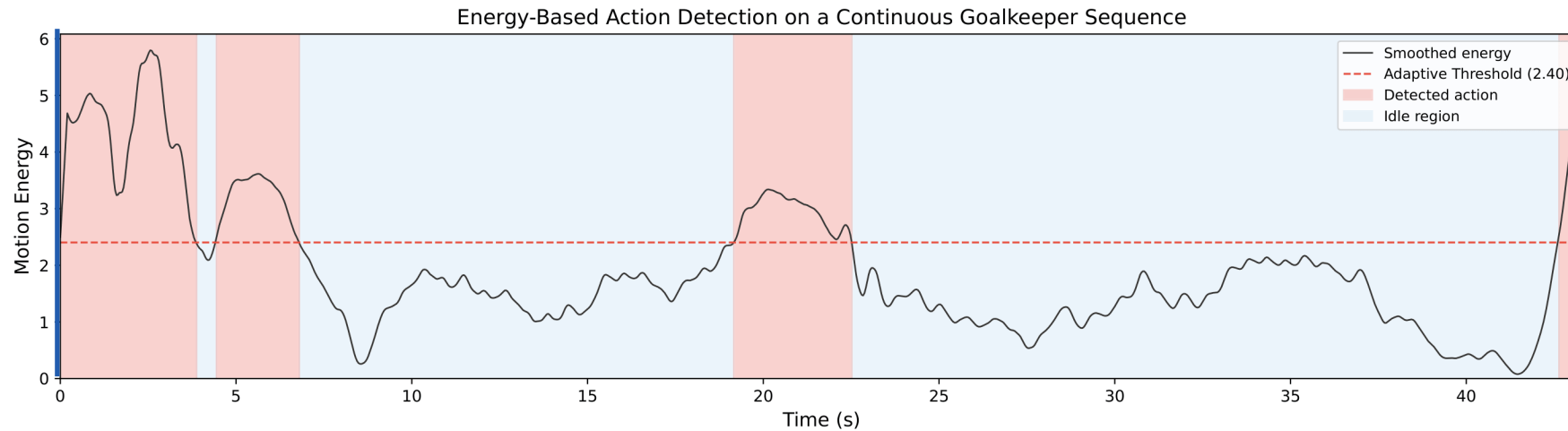
Average bone lengths:

Neck to Mid hip: 0.6m
Left hip to Mid hip : 0.113m
Right hip to Mid hip : 0.113m
Neck to Left shoulder: 0.2m

...



Data Preprocessing — Action extraction



Motion energy: total movement of the whole skeletal structure relative to the previous frame

$$E_t = \sum_{v=1}^V ||\mathbf{p}_{t,v} - \mathbf{p}_{t-1,v}||_2, \text{ where } \mathbf{p}_{t,v} \text{ is 3D position of joint } v \text{ at time } t$$

baseline “non-action” energy

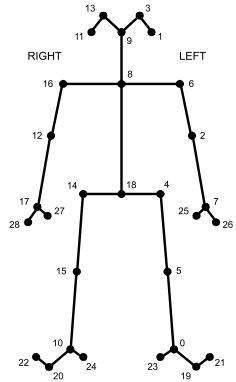
$$\text{MAD}(E) = \text{Median}_{t=1,\dots,T}(|E_t - \text{Median}_{t=1,\dots,T}(E_t)|)$$

Usually, how much does the motion energy **fluctuate** from the **baseline**?

$$\text{Threshold} = \text{Median}_{t=1,\dots,T}(E_t) + \lambda \cdot \text{MAD}(E)$$

At what **intensity** does movement **count as an action**?

Representation Learning

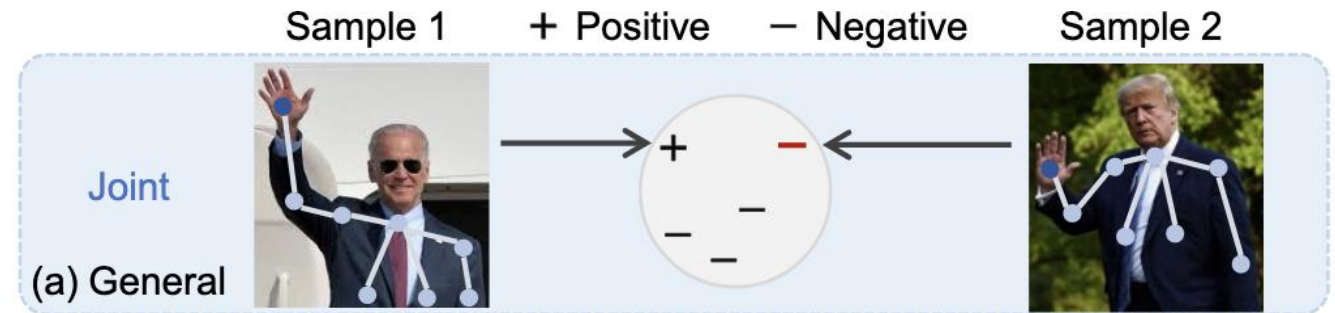


*How to learn the weights for this encoder, in a **self-supervised** manner?*

Contrastive Learning Framework

- In **traditional self-supervised contrastive learning frameworks**, a **positive sample** for each input is created using *data augmentation*, whereas all other inputs are treated as **negative samples**.

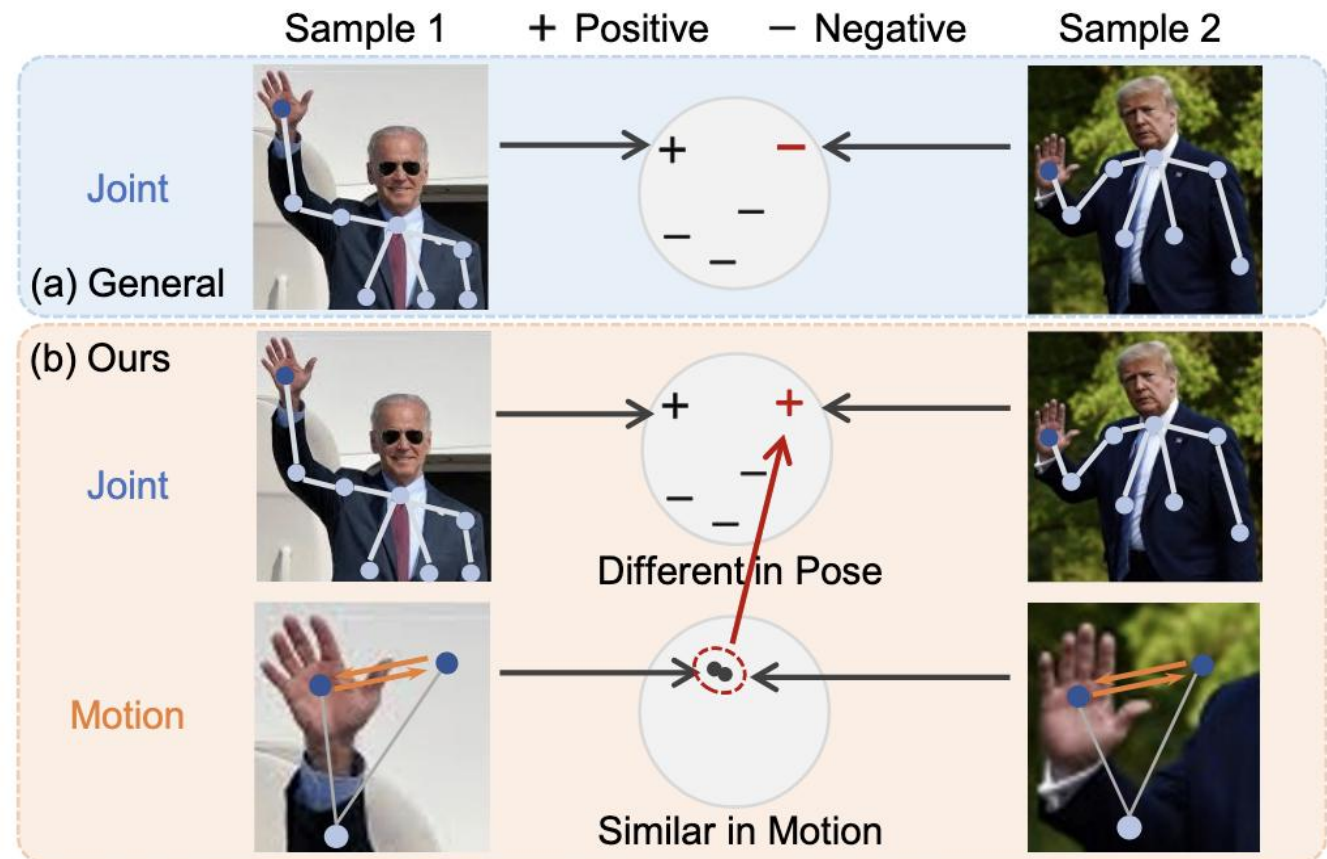
Problem: You can assign two similar motion pairs as negative, pushing them apart in the representation space, leading to *an unreasonable learned space*.



Li, L., Wang, M., Ni, B., Wang, H., Yang, J., & Zhang, W. (2021). 3D Human Action Representation Learning via Cross-View Consistency Pursuit.

Cross-view Contrastive Learning Framework

- In a **cross-view contrastive learning framework**, similarity information is inferred from other, *easily computable* views. If two samples are highly similar in one view, they are treated as positive samples across all views.



Li, L., Wang, M., Ni, B., Wang, H., Yang, J., & Zhang, W. (2021). 3D Human Action Representation Learning via Cross-View Consistency Pursuit.

Unsupervised Action Discovery



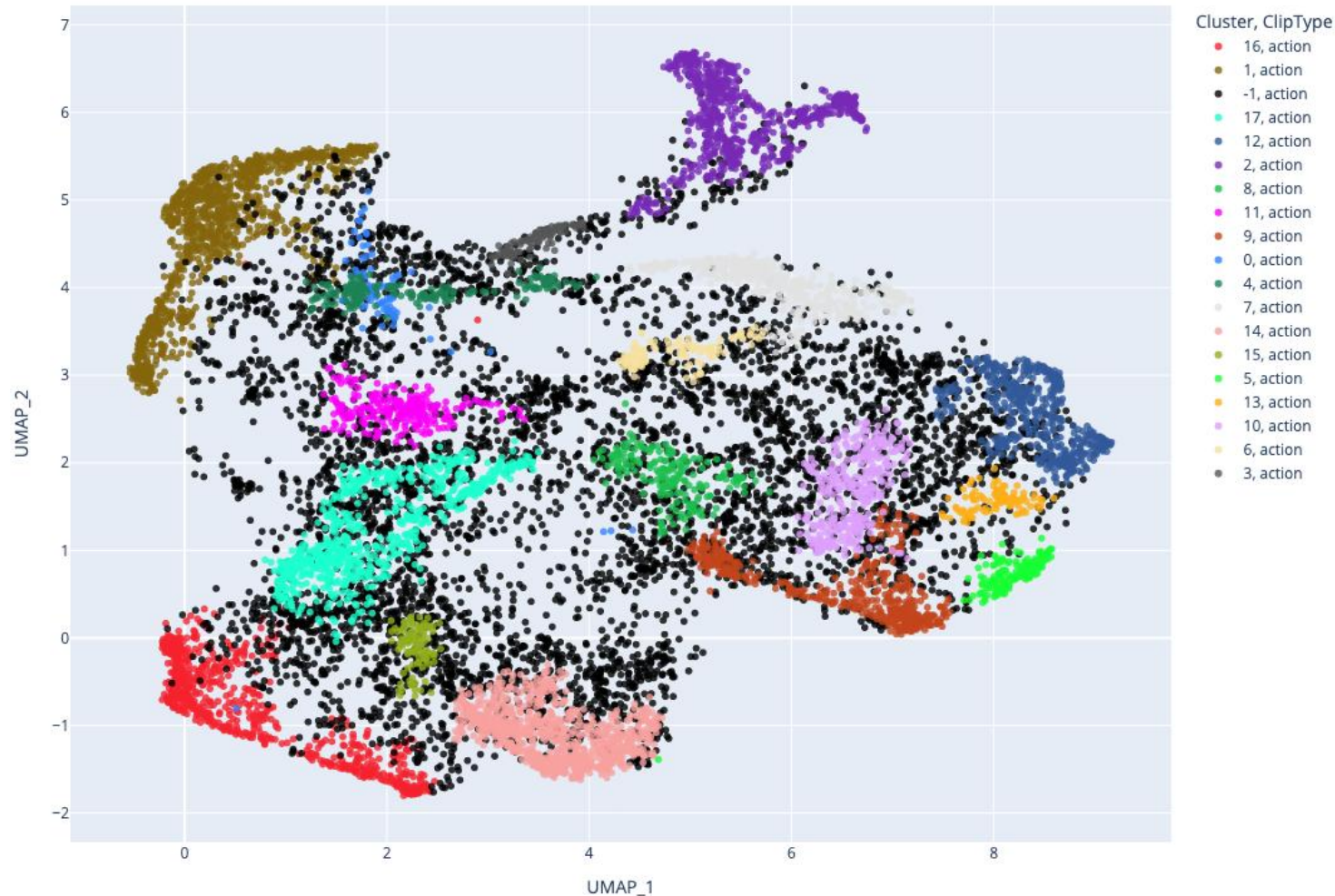
Problem: Clustering in high dimension is difficult due to *Curse of Dimensionality*

Solution: We first reduce the dimensionality of the learned *128-dimensional embedding space to 20 dimensions* using **UMAP** (Uniform Manifold Approximation and Projection).

We apply **HDBSCAN** on the reduced space, grouping similar representations through *density-based clustering*.

Visualisation of the Semantic Structure in the Learned Space

Action-Only Clusters (Custom: NN=80, MCS=60, MS=20)

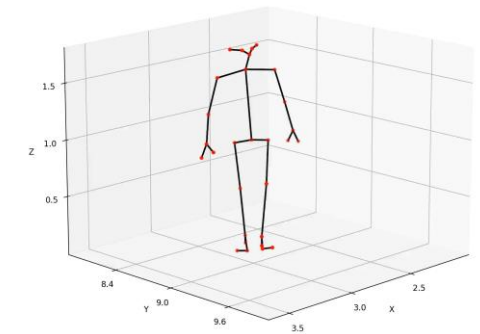
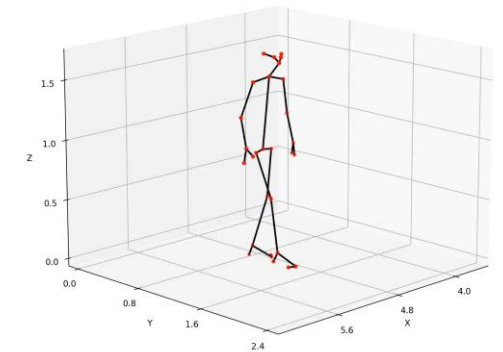


Cluster ID	Type
0	Walking Around
1	Walking Around
2	Kick the Ball
3	Pass / Throw
4	Bending Down
5	Walk + Grab
6	Ready Pose
7	Saves/Dives
8	Receive Pass
9	Hand Communication
10	Pass + Move
11	Ready + Shuffle
12	Sprint + Grab
13	Pass + Move
14	Moving Back
15	Shuffle + Ready
16	Hand Signals / Stretch
17	Ready Pose

Characteristics of the Clusters

Cluster 2 and 3 involve *kicks* and *passes*, and this has high ankle velocity and high ankle to hip velocity ratio, indicating isolated movement of the ankles.

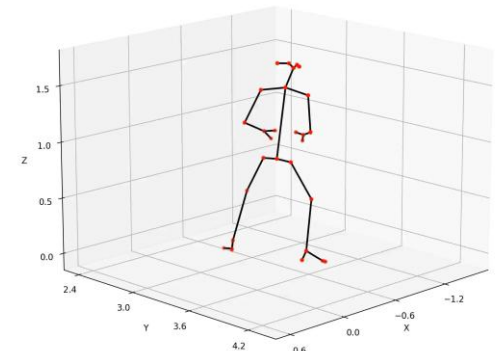
Cl.	Description	N	Disp (m)	Wrist Min (m)	Foot Max (m)	Hip Vel (m/s)	Ank Vel (m/s)	Wrist Vel (m/s)	W/H Ratio	A/H Ratio
-1	Noise	5274	4.52 ± 2.98	0.736 ± 0.214	0.343 ± 0.201	5.775 ± 3.262	11.499 ± 5.771	8.997 ± 5.280	2.546 ± 3.111	2.803 ± 2.257
0	Walking Around	78	3.139 ± 0.812	0.871 ± 0.135	0.243 ± 0.058	3.235 ± 0.501	7.166 ± 1.046	5.064 ± 1.158	2.619 ± 1.666	3.050 ± 0.954
1	Walking Around	1179	3.212 ± 1.117	0.900 ± 0.123	0.272 ± 0.101	2.512 ± 0.816	8.827 ± 1.811	4.323 ± 1.706	2.476 ± 1.515	3.193 ± 0.980
2	Kick the ball	761	3.652 ± 1.785	0.732 ± 0.114	0.644 ± 0.179	7.189 ± 2.155	17.897 ± 4.353	11.146 ± 3.735	3.540 ± 1.555	5.262 ± 2.846
3	Pass / throw	98	1.596 ± 0.929	0.762 ± 0.190	0.556 ± 0.261	3.964 ± 1.762	12.149 ± 3.891	8.195 ± 4.390	4.223 ± 2.749	6.640 ± 4.988
4	Bending down	199	1.529 ± 0.943	0.562 ± 0.303	0.379 ± 0.274	3.319 ± 2.426	8.149 ± 5.189	6.564 ± 3.616	4.768 ± 4.306	5.518 ± 6.415
5	Walk + grab	196	8.947 ± 1.990	0.714 ± 0.249	0.318 ± 0.138	8.161 ± 2.377	16.094 ± 4.218	13.473 ± 4.512	2.656 ± 4.692	2.672 ± 1.295
6	Ready for shot	118	3.675 ± 1.459	0.584 ± 0.153	0.331 ± 0.101	7.470 ± 2.112	14.751 ± 3.432	11.762 ± 3.506	2.282 ± 1.288	2.186 ± 0.397
7	Saves / Dives	490	3.653 ± 2.593	<u>0.058 ± 0.267</u>	0.723 ± 0.315	9.386 ± 4.745	16.492 ± 7.216	16.755 ± 6.391	3.357 ± 3.995	3.709 ± 5.787
8	Receive pass	197	4.548 ± 1.863	0.797 ± 0.127	0.391 ± 0.128	5.370 ± 1.331	11.183 ± 2.683	7.669 ± 2.023	1.912 ± 1.170	2.953 ± 2.342
9	Hand comm.	489	5.178 ± 1.456	0.785 ± 0.225	0.285 ± 0.111	4.872 ± 1.317	10.385 ± 2.469	8.334 ± 2.798	3.366 ± 5.818	2.975 ± 1.339
10	Pass + move	368	7.777 ± 1.801	0.804 ± 0.157	0.486 ± 0.122	7.503 ± 1.938	13.807 ± 3.813	10.267 ± 2.918	<u>1.673 ± 0.786</u>	2.404 ± 1.753
11	Ready + shuffle	259	1.821 ± 0.991	0.686 ± 0.127	0.236 ± 0.068	3.966 ± 1.087	8.510 ± 2.137	5.836 ± 1.894	1.990 ± 1.025	2.538 ± 0.504
12	Sprint + grab	541	14.879 ± 4.798	0.613 ± 0.256	0.591 ± 0.162	14.757 ± 3.816	24.235 ± 6.335	20.561 ± 5.988	1.765 ± 0.953	<u>1.944 ± 1.967</u>
13	Pass + move	112	10.457 ± 1.792	0.777 ± 0.202	0.472 ± 0.111	8.706 ± 1.578	15.608 ± 3.593	11.882 ± 2.592	1.818 ± 2.090	2.783 ± 3.357
14	Moving back	855	7.772 ± 2.575	0.788 ± 0.157	0.270 ± 0.106	7.324 ± 2.398	12.924 ± 3.991	9.894 ± 3.684	1.766 ± 2.616	2.027 ± 0.880
15	Shuffle + ready	109	4.054 ± 0.793	0.772 ± 0.116	0.182 ± 0.065	4.291 ± 0.804	7.928 ± 1.375	5.663 ± 1.391	1.850 ± 1.350	2.235 ± 0.388
16	Hand signals / stretch	730	3.485 ± 1.414	0.885 ± 0.119	0.237 ± 0.091	3.108 ± 1.196	6.916 ± 2.464	5.337 ± 2.310	2.835 ± 2.469	3.338 ± 2.888
17	Ready pose	743	2.051 ± 1.057	0.681 ± 0.199	0.201 ± 0.137	3.573 ± 1.367	7.346 ± 2.612	5.293 ± 2.814	2.086 ± 2.117	2.514 ± 0.642



Characteristics of the Clusters

Cluster 7 involves *saves* and *dives*, and this has high wrist velocity, along with maximum foot height highlighting jumps and quick movements of the wrist.

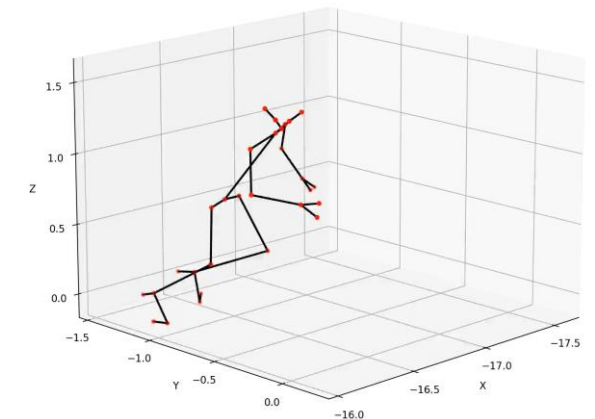
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4	Bending down	199	1.529 ± 0.943	0.562 ± 0.303	0.379 ± 0.274	3.319 ± 2.420	8.148 ± 5.188	6.564 ± 3.616	4.768 ± 4.305	5.512 ± 6.417
5	Walk + grab	196	8.947 ± 1.990	0.714 ± 0.249	0.318 ± 0.138	8.161 ± 2.377	16.094 ± 4.218	13.473 ± 4.512	2.656 ± 4.692	2.672 ± 1.295
6	Ready for shot	118	3.675 ± 1.459	0.584 ± 0.153	0.221 ± 0.101	7.470 ± 2.112	14.751 ± 3.432	11.763 ± 3.506	2.282 ± 1.288	2.186 ± 0.397
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Characteristics of the Clusters

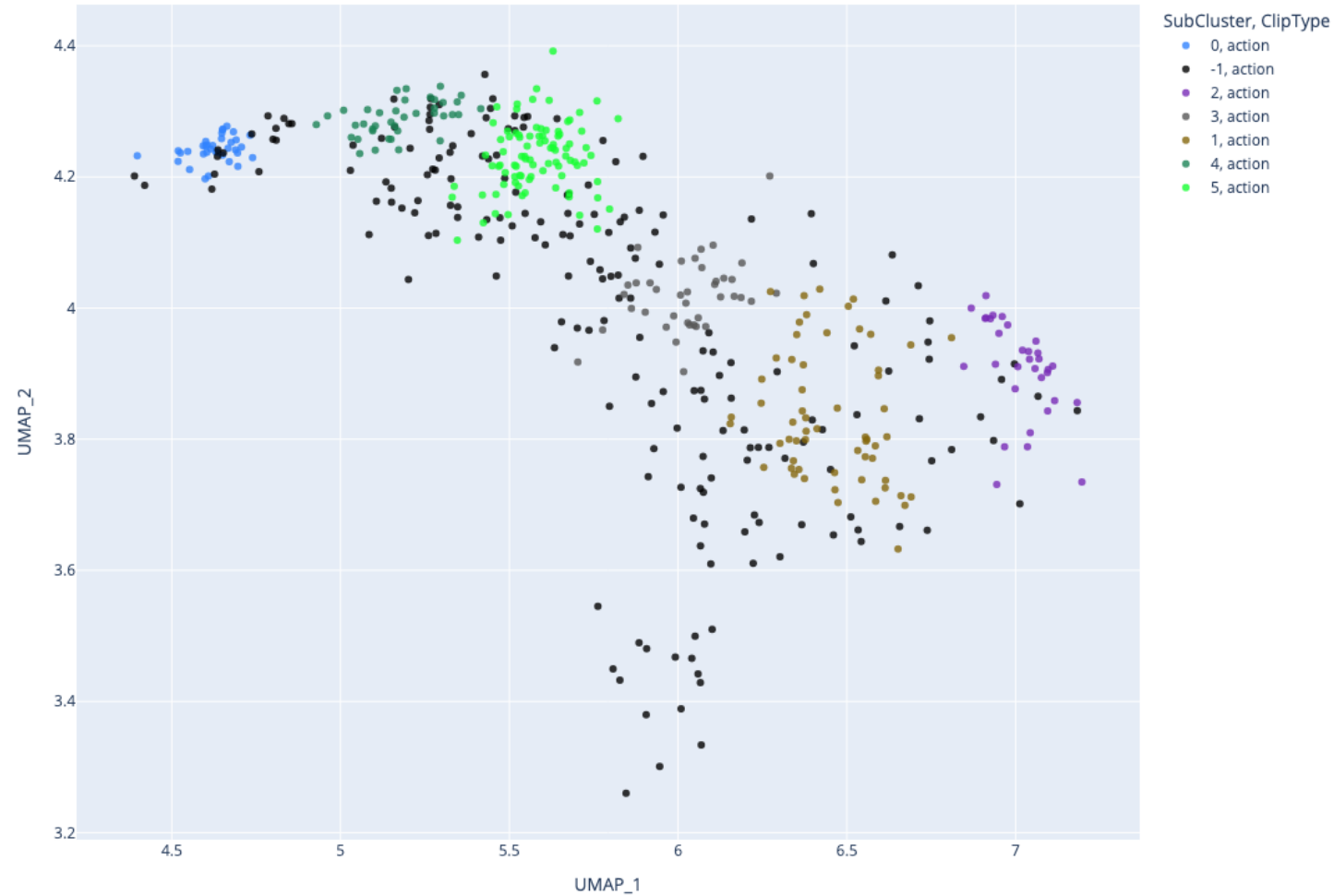
Cluster 12 corresponds to *sprinting*, characterised by high hip displacement, hip velocity, and ankle velocity, reflecting the rapid and coordinated lower-body mechanics associated with this action.

Cl.	Description	N	Disp (m)	Wrist Min (m)	Foot Max (m)	Hip Vel (m/s)	Ank Vel (m/s)	Wrist Vel (m/s)	W/H Ratio	A/H Ratio
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14	Moving back	855	7.772 ± 2.575	0.788 ± 0.157	0.270 ± 0.106	7.324 ± 2.398	12.924 ± 3.991	9.894 ± 3.684	1.766 ± 2.616	2.027 ± 0.880
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Semantic Structure within Saves

Sub-clusters of Cluster 7 (NN=30, MCS=30, MS=5)

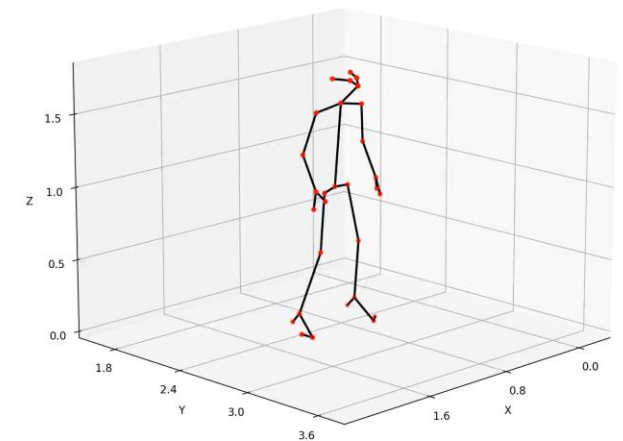


Cluster ID	Type
0	Noisy Samples
1	Jumps and Punches
2	Drop Low Saves
3	Mixed
4	Dive Forward to Grab Ball
5	Lateral Dives

Characteristics of Save Subclusters: Jumps and Punches

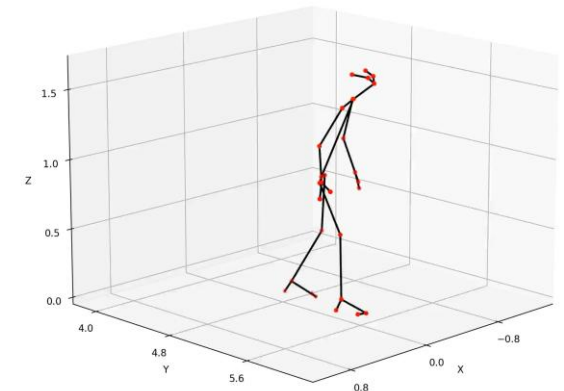
Sub-Cluster	Description	N	Min Neck	Max Neck	Min Hip
sub-0	Noisy Samples	35	0.42 ± 0.43	1.03 ± 0.45	0.18 ± 0.26
sub-1	Jumps and Punches	59	0.48 ± 0.34	1.70 ± 0.22	0.21 ± 0.18
sub-2	Drop Low Saves	32	0.35 ± 0.26	1.49 ± 0.16	0.16 ± 0.13
sub-3	Mixed	39	0.32 ± 0.25	1.58 ± 0.15	0.15 ± 0.11
sub-4	Dive Forward to Grab Ball	39	0.21 ± 0.19	1.42 ± 0.15	0.20 ± 0.15
sub-5	Lateral Dives	89	0.23 ± 0.15	1.51 ± 0.17	0.14 ± 0.06

Sub-Cluster	Description	N	Body Angle (deg)	Maximal Extension (m)
sub-0	Noisy Samples	35	37.69 ± 30.43	1.53 ± 0.27
sub-1	Jumps and Punches	59	60.27 ± 21.49	1.76 ± 0.16
sub-2	Drop Low Saves	32	46.29 ± 20.98	1.65 ± 0.14
sub-3	Mixed	39	39.78 ± 27.81	1.81 ± 0.16
sub-4	Dive Forward to Grab Ball	39	33.77 ± 25.22	1.60 ± 0.23
sub-5	Lateral Dives	89	33.85 ± 22.10	1.86 ± 0.13

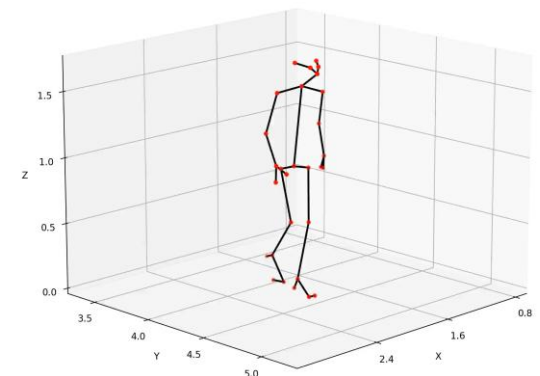


Characteristics of Save Subclusters: Drop Low Saves

Sub-Cluster	Description	N	Min Neck	Max Neck	Min Hip
sub-0	Noisy Samples	35	0.42 ± 0.43	1.03 ± 0.45	0.18 ± 0.26
sub-1	Jumps and Punches	59	0.48 ± 0.34	1.70 ± 0.22	0.21 ± 0.18
sub-2	Drop Low Saves	32	0.35 ± 0.26	1.49 ± 0.16	0.16 ± 0.13
sub-3	Mixed	39	0.32 ± 0.25	1.58 ± 0.15	0.15 ± 0.11
sub-4	Dive Forward to Grab Ball	39	0.21 ± 0.19	1.42 ± 0.15	0.20 ± 0.15
sub-5	Lateral Dives	89	0.23 ± 0.15	1.51 ± 0.17	0.14 ± 0.06



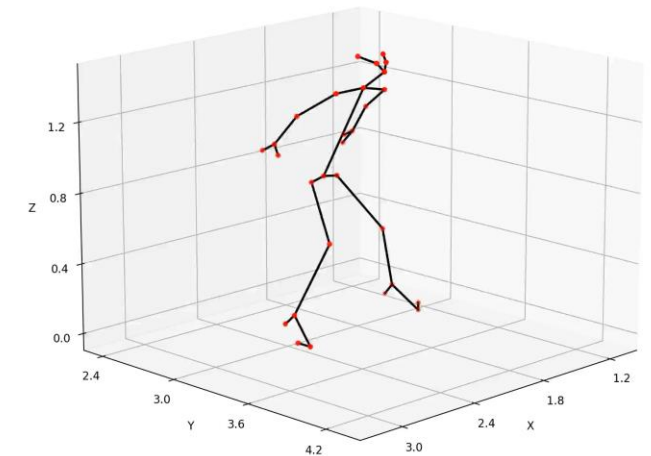
Sub-Cluster	Description	N	Body Angle (deg)	Maximal Extension (m)
sub-0	Noisy Samples	35	37.69 ± 30.43	1.53 ± 0.27
sub-1	Jumps and Punches	59	60.27 ± 21.49	1.76 ± 0.16
sub-2	Drop Low Saves	32	46.29 ± 20.98	1.65 ± 0.14
sub-3	Mixed	39	39.78 ± 27.81	1.81 ± 0.16
sub-4	Dive Forward to Grab Ball	39	33.77 ± 25.22	1.60 ± 0.23
sub-5	Lateral Dives	89	33.85 ± 22.10	1.86 ± 0.13



Characteristics of Save Subclusters: Dive Forward to Grab Ball

Sub-Cluster	Description	N	Min Neck	Max Neck	Min Hip
sub-0	Noisy Samples	35	0.42 ± 0.43	1.03 ± 0.45	0.18 ± 0.26
sub-1	Jumps and Punches	59	0.48 ± 0.34	1.70 ± 0.22	0.21 ± 0.18
sub-2	Drop Low Saves	32	0.35 ± 0.26	1.49 ± 0.16	0.16 ± 0.13
sub-3	Mixed	39	0.32 ± 0.25	1.58 ± 0.15	0.15 ± 0.11
sub-4	Dive Forward to Grab Ball	39	0.21 ± 0.19	1.42 ± 0.15	0.20 ± 0.15
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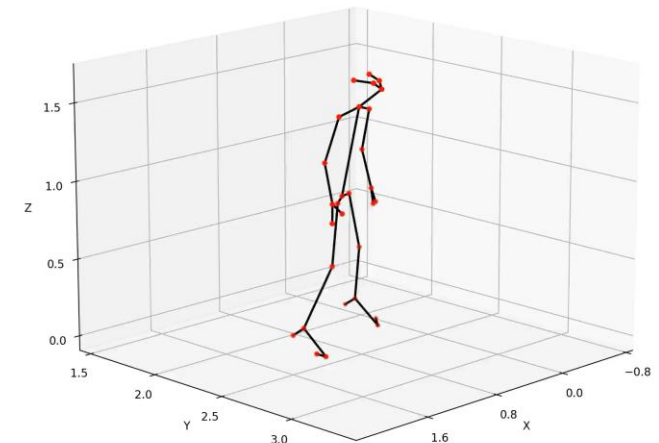
Sub-Cluster	Description	N	Body Angle (deg)	Maximal Extension (m)
sub-0	Noisy Samples	35	37.69 ± 30.43	1.53 ± 0.27
sub-1	Jumps and Punches	59	60.27 ± 21.49	1.76 ± 0.16
sub-2	Drop Low Saves	32	46.29 ± 20.98	1.65 ± 0.14
sub-3	Mixed	39	39.78 ± 27.81	1.81 ± 0.16
sub-4	Dive Forward to Grab Ball	39	33.77 ± 25.22	1.60 ± 0.23
sub-5	Lateral Dives	89	33.85 ± 22.10	1.86 ± 0.13



Characteristics of Save Subclusters: Lateral Dives

Sub-Cluster	Description	N	Min Neck	Max Neck	Min Hip
sub-0	Noisy Samples	35	0.42 ± 0.43	1.03 ± 0.45	0.18 ± 0.26
sub-1	Jumps and Punches	59	0.48 ± 0.34	1.70 ± 0.22	0.21 ± 0.18
sub-2	Drop Low Saves	32	0.35 ± 0.26	1.49 ± 0.16	0.16 ± 0.13
sub-3	Mixed	39	0.32 ± 0.25	1.58 ± 0.15	0.15 ± 0.11
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Conclusion

These *biomechanical metrics* are consistent with our labels through visual inspection, highlighting that our model learnt meaningful representations. These representations can directly enhance downstream prediction and classification tasks, offering a practical tool for goalkeeping coaches and opening new directions for data-driven athlete development.

Some ideas for future downstream tasks:

1. Save type *classification* (need for labelled data)
2. Integrate into *xG models* to include goalkeeper skeletal context
3. *Retrieval* of similar action sequences
4. Comparing keeper profiles for *recruitment* (compare distributions of action sequences)

Thank you for your attention!



Scan to read the report!

Discussion

Extra Slides

Prior Work

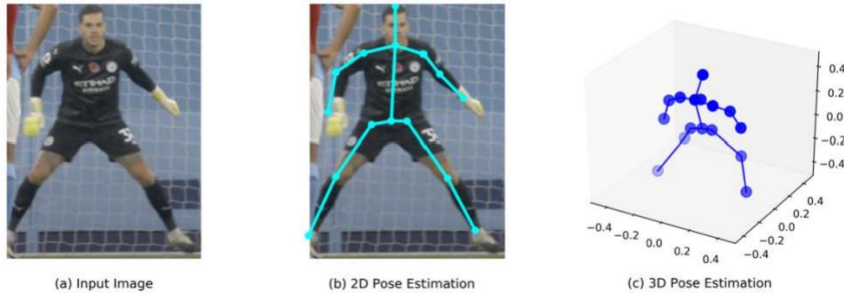


Illustration from Wear et al. "Learning from the Pros: Extracting Professional Goalkeeper Technique from Broadcast Footage" 2022 MIT Sloan Sports Analytics Conference

⚠️ STATIC AND HAND-CRAFTED FEATURES

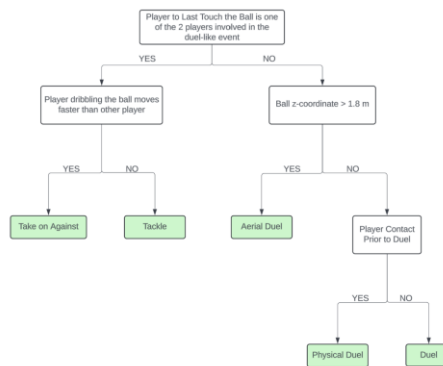


Illustration from G. C. Bian, "Soccer last touch and automatic event detection with skeletal tracking data," 2023, Master's thesis.

⚠️ PREDEFINED EVENT TAXONOMY

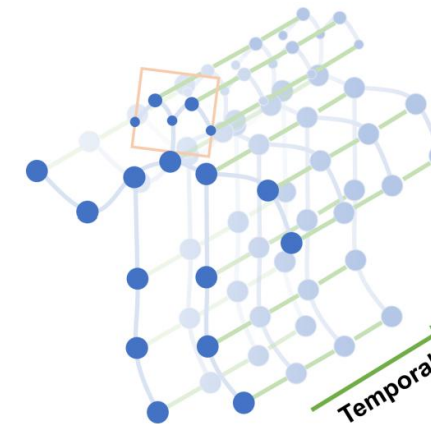


Illustration from Yan et al. "Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition" 2018 AAAI

⚠️ SUPERVISED

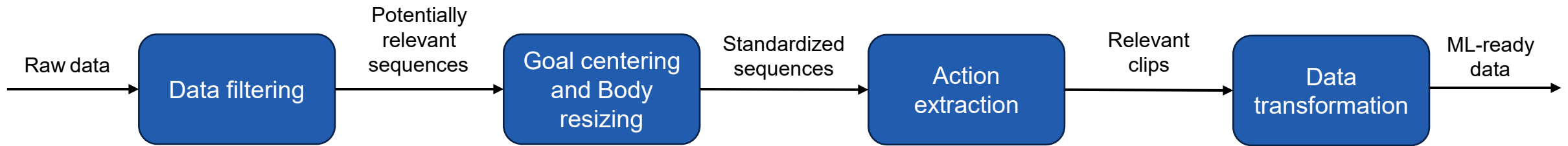


Illustration from Yeung et al., "AutoSoccerPose: Automated 3D posture Analysis of Soccer Shot Movements," 2024, CVPR

⚠️ ONE ACTION CLASS AND FIXED NUMBER OF CLUSTERS

UNSUPERVISED, TEMPORAL AND LEARNABLE ACTION TAXONOMY :)

Data Preprocessing — Overview



Dataset

Raw skeletal data example:

```
{  
  "joints": [  
    {  
      "lAnkle": [20.104230880737305, 2.6160707473754883, 0.26433616876602173],  
      "lEar": [19.9064998626709, 2.545010566711426, 1.6722733974456787],  
      ...  
      "time": 8.01838  
    }  
  ],  
  "personId": {"heId": "aeeff2d2-7141-419f-742d-08d958b7d325", "uefaId": "250085369"},  
  "role": {"id": 1, "name": "Outfielder"},  
  "trackId": 17701  
}
```

ONE FRAME

Data Preprocessing — Tensor representation

- 5D tensor of shape (N, C, T, V, M) where:
 - N number of sequences
 - $C = 3$ spatial coordinates in 3D (x, y, z)
 - $T = 50$ fixed temporal length (number of frames) per sequence
 - V number of skeletal joints
 - M number of subjects included in the sample
- Approximately 80-20 train-validation split using teams eliminated in group stage
- Median shoulder-vector alignment and root centering
- Two different representations:
 - Native: 29 joints, 25 fps, regular coordinates, $M = 1$
 - 28 285 samples, 17 103 active, 11 182 idle
 - NTU-aligned: 25 joints, 30 fps, Kinect v2 coordinates, $M = 2$
 - 24 369 samples, 18 307 active, 10 776 idle

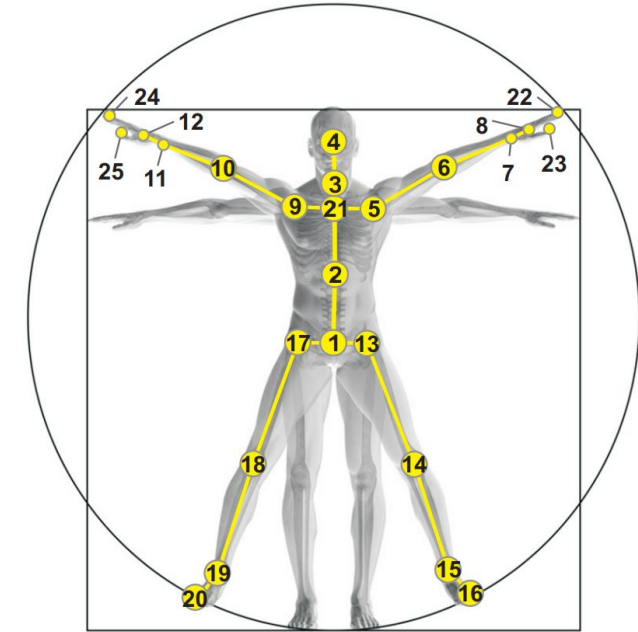
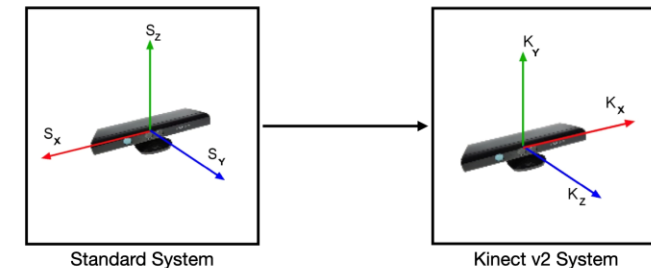


Illustration from Shahroudy et al. "NTU RGB+D: A Large Scale Dataset for 3D Human Activity Analysis," 2016 CVPR

- | | | | |
|-------------------------|--------------------|------------------|----------------------------|
| 1 - Base of the spine | 8 - Left hand | 15 - Left ankle | 22 - Tip of the left hand |
| 2 - Middle of the spine | 9 - Right shoulder | 16 - Left foot | 23 - Left thumb |
| 3 - Neck | 10 - Right elbow | 17 - Right hip | 24 - Tip of the right hand |
| 4 - Head | 11 - Right wrist | 18 - Right knee | 25 - Right thumb |
| 5 - Left shoulder | 12 - Right hand | 19 - Right ankle | |
| 6 - Left elbow | 13 - Left hip | 20 - Right foot | |
| 7 - Left wrist | 14 - Left knee | 21 - Spine | |



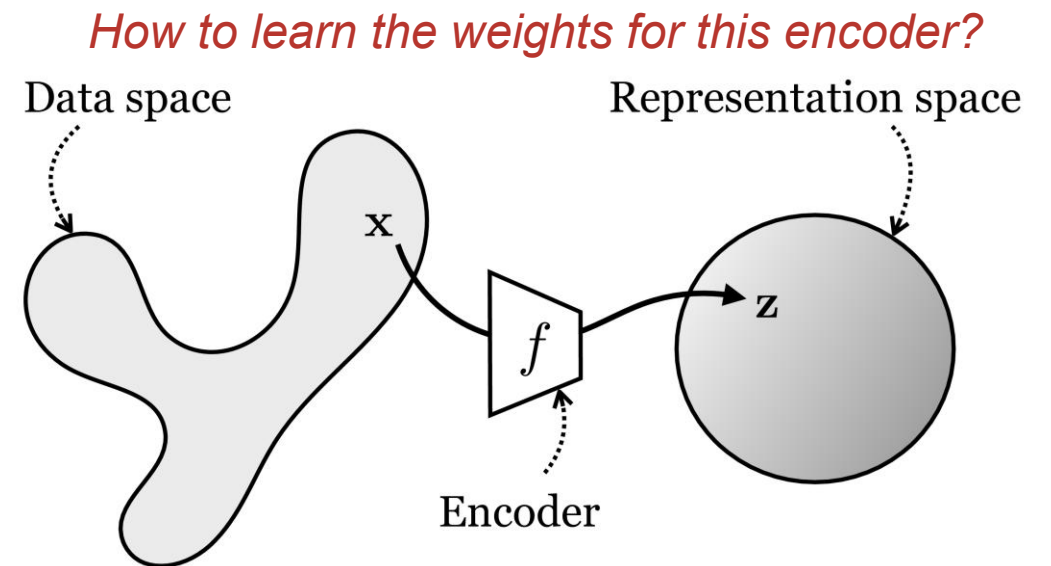
Representation Learning

"learning representations of the data that make it easier to extract useful information when building classifiers or other predictors"

Y. Bengio, A. Courville and P. Vincent, "Representation Learning: A Review and New Perspectives," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 35, no. 8, pp. 1798-1828, Aug. 2013, doi: 10.1109/TPAMI.2013.50.

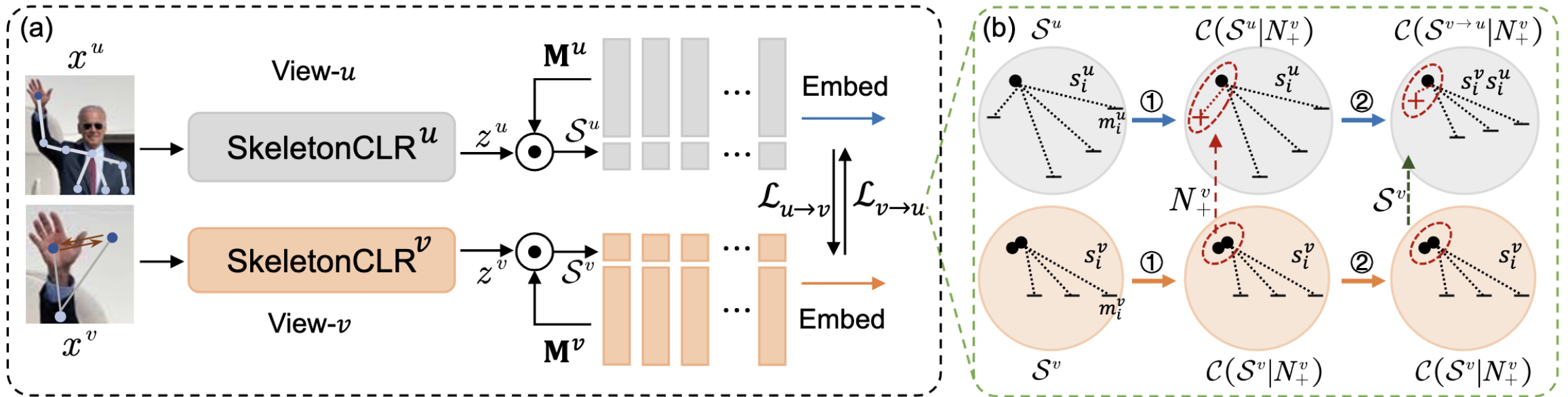
Goal: Learn meaningful representations for each goalkeeper sequence in a high-dimensional space, and then this makes it easier to *characterise* each sequence in this learned space

Solution: To learn these representations in a *self-supervised* manner, we use the **contrastive learning framework**.



Model Architecture

- To learn the representations, we use the **CrosSCLR** (*Cross Skeleton Contrastive Learning Representation*) model architecture. The encoder is implemented using an ST-GCN.



Li, L., Wang, M., Ni, B., Wang, H., Yang, J., & Zhang, W. (2021). 3D Human Action Representation Learning via Cross-View Consistency Pursuit.

Data Augmentations

1. Temporal Crop
2. Shear
3. Mirroring (applied with a probability of **0.5**)
4. Gaussian Noise

They are all applied *sequentially* to each input sample.

Definition of Biomechanical Characteristics for Body Joints (1/2)

1. **Ankle Velocity:** The maximum speed recorded for the faster of the two ankles in the motion sequence.
2. **Hip Velocity:** The highest instantaneous speed, measured in meters per second (m/s), reached by the mid-hip joint in the motion sequence.
3. **Wrist Velocity:** The maximum speed recorded for the faster of the two wrists in the motion sequence.
4. **Wrist to Hip Velocity Ratio:** The ratio of wrist velocity to hip velocity, calculated specifically at the *frame where the wrist is moving at its fastest*.
5. **Ankle to Hip Velocity Ratio:** The ratio of ankle velocity to hip velocity, calculated at the frame of peak ankle speed. A higher ratio typically indicates a more explosive leg snap relative to forward body travel.

Definition of Biomechanical Characteristics for Body Joints (2/2)

1. **Wrist Min:** The lowest vertical point (z-coordinate) reached by either the left or right wrist during the entire sequence.
2. **Foot Max:** The highest vertical point (z-coordinate) reached by either ankle joint.

Definition of Biomechanical Characteristics for Saves

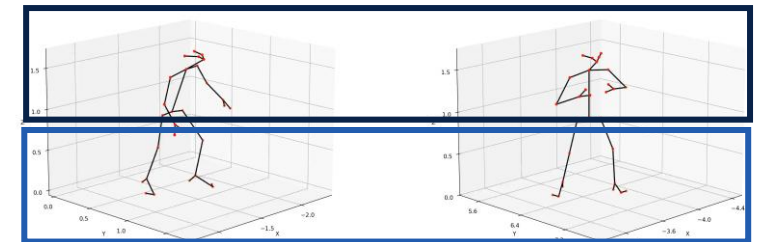
1. **Min/Max Neck:** The lowest and highest vertical positions of the neck joint recorded during the movement sequence, averaged across all clips in the cluster.
2. **Min Hip:** The lowest vertical position reached by the mid-hip at any point during the sequence.
3. **Maximal Extension:** It is the maximum straight-line (Euclidean) distance between any wrist joint and any ankle joint at the specific frame where this distance is greatest. (proxy for when the keeper is diving)
4. **Body Angle (deg):** The lean of the player's torso relative to the ground. It is calculated as the angle between the ground plane and the vector (line) connecting the mid-hip to the neck at the frame of maximal extension.

Validation of the Clusters

1. We randomly sampled **20** samples from the validation set (not seen during training).
2. For each sample, we retrieved *5 of its nearest neighbours* in the embedding space from the training set to label it.
3. Then we verified if the assigned label was accurate through *visual inspection*.
4. Of the 20 samples, 12 were assigned a non-noise label — 11 of which were verified as correct through visual inspection (**92%** accuracy).

Future Work

1. **Incorporate Game Context:** Enhance initial feature sets by integrating *broader game context* alongside joint data.
2. **Validation of Cluster Quality:** Hyperparameter was tuned by visual inspection of clusters, which is limited by human resources. A more systematic approach needs to be explored.
3. **Improve Data Filtering:** Both Action Extraction and Spatial Filtering use hard thresholds, making abrupt cuts to the sequence.
4. **Improved Data Augmentations:** Include more goalkeeper action specific augmentations as the current set is limited.
5. **Separate Data Streams:** Introduce separate data streams for upper and lower body, and fusing them later in the pipeline. This enforces the model to learn both upper and lower body representations independently before combining them, reducing the risk of one body region dominating the learned features.



More on Future Work

6. **Improved Data Preprocessing Pipeline:** In our data preprocessing pipeline, we anchor the hip to the origin, hiding the actual movement of the skeleton away from the model. One could perform ablation studies on this and different alignment strategies to see its effect on model performance.